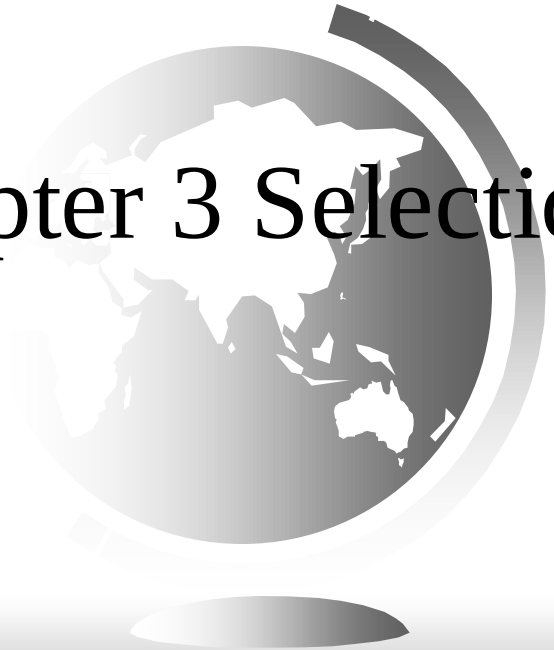


Chapter 3 Selections

Liang, Introduction to Java Programming and Data Structures, Twelfth Edition, Global Edition,
Copyright © 2022 Pearson Education Ltd. All rights reserved.

Motivations

If you assigned a negative value for radius in the program, *ComputeAreaWithConsoleInput.java*, the program would print an invalid result.

If the radius is negative, you don't want the program to compute the area. How can you deal with this situation?



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Outline

- boolean type and operators
- Selection Control
  - one-way if statements
  - two-way if-else statements
  - multi-way if statements.
- Logical Operators
- switch statements
- Conditional expression
- Mathematical Functions



Copyright © 2022 Pearson Education Ltd. All rights reserved.

The `boolean` Type and Operators

- Often in a program you need to compare two values, such as whether *i* is greater than *j*.

- Java provides six comparison operators (also known as relational operators) that can be used to compare two values.

- The result of the comparison is a Boolean value: true or false.

  - **`boolean b = (1 > 2);`**



Copyright © 2022 Pearson Education Ltd. All rights reserved.

The `boolean` Type and Operators

- The result of the comparison is a Boolean value: **true** or **false**.

```
boolean b = (1 > 2);
boolean lightsOn = true;
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Relational Operators

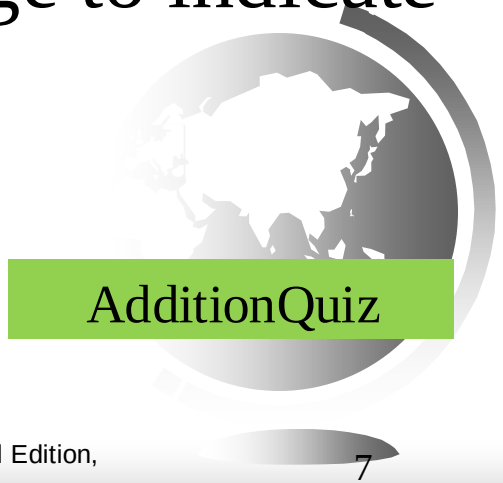
Java Operator	Mathematics Symbol	Name	Example (radius is 5)	Result
`<`	<	less than	`radius < 0`	`false`
`<=`	≤	less than or equal to	`radius <= 0`	`false`
`>`	>	greater than	`radius > 0`	`true`
`>=`	≥	greater than or equal to	`radius >= 0`	`true`
`==`	=	equal to	`radius == 0`	`false`
`!=`	≠	not equal to	`radius != 0`	`true`



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: A Simple Math Learning Tool

This example creates a program to let a first grader practice additions. The program randomly generates two single-digit integers number1 and number2 and displays a question such as “What is 7 + 9?” to the student. After the student types the answer, the program displays a message to indicate whether the answer is true or false.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

if Statements

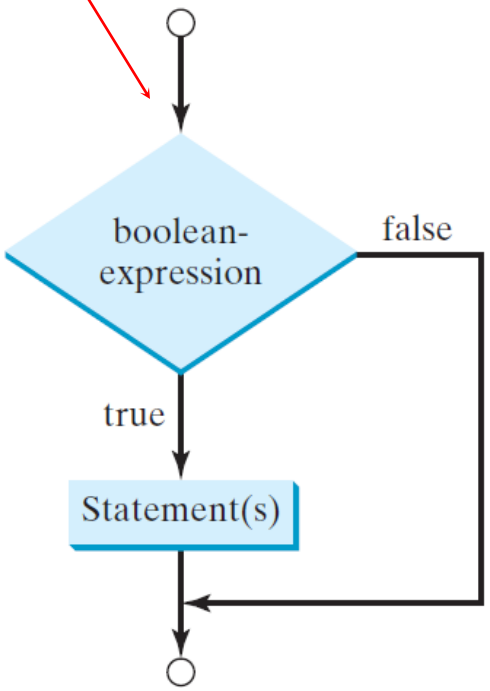
- An **if** statement is a construct that enables a program to specify alternative paths of execution.



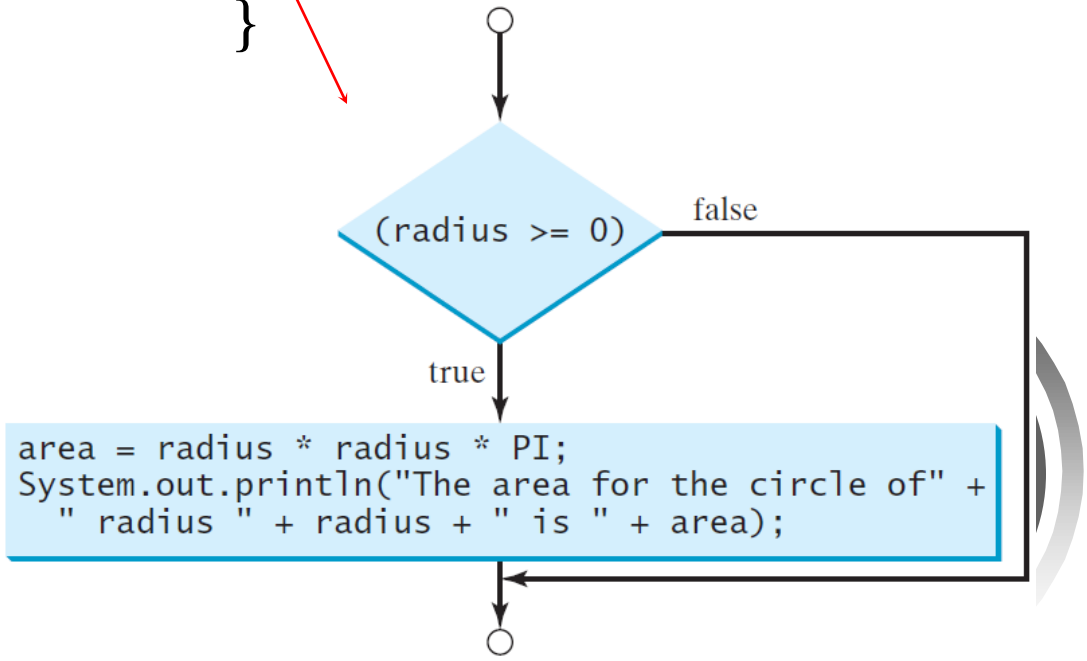
Copyright © 2022 Pearson Education Ltd. All rights reserved.

One-way `if` Statements

```
if (boolean-expression) {
  statement(s);
}
```




```
if (radius >= 0) {
  area = radius * radius * PI;
  System.out.println("The area"
    + " for the circle of radius "
    + radius + " is " + area);
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Note

```
if i > 0 {
  System.out.println("i is positive");
}
```

(a) Wrong

```
if (i > 0) {
  System.out.println("i is positive");
}
```

(b) Correct

```
if (i > 0) {
  System.out.println("i is positive");
}
```

(a)

Equivalent

```
if (i > 0)
  System.out.println("i is positive");
```

(b)



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Simple if Demo

Write a program that prompts the user to enter an integer. If the number is a multiple of 5, print HiFive. If the number is divisible by 2, print HiEven.

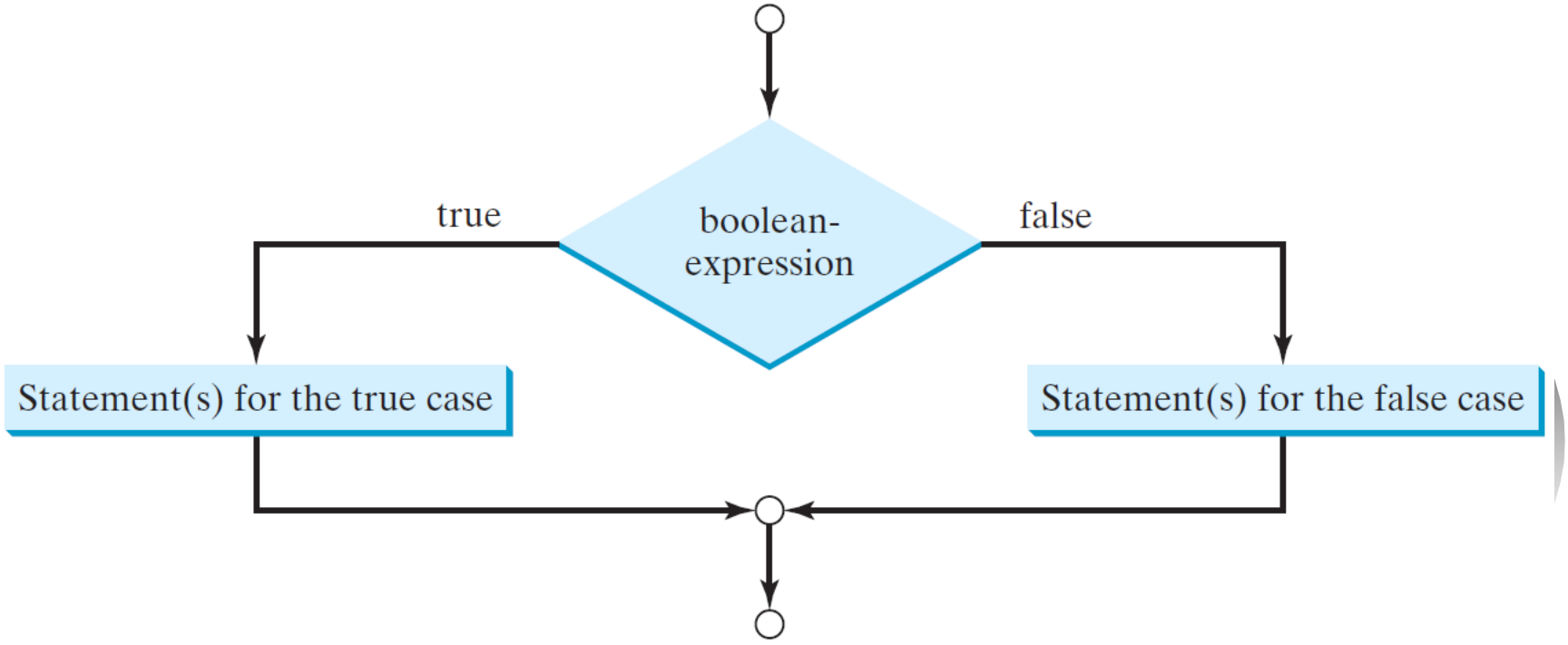
SimpleIfDemo



Copyright © 2022 Pearson Education Ltd. All rights reserved.

The Two-way `if` Statement

```
if (boolean-expression) {
  statement(s)-for-the-true-case;
}
else {
  statement(s)-for-the-false-case;
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

if-else Example

```
if (radius >= 0) {
  area = radius * radius * 3.14159;

  System.out.println("The area for the “
    + “circle of radius " + radius +
    " is " + area);
}
else {
  System.out.println("Negative input");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Multiple Alternative if Statements

```
if (score >= 90.0)
  System.out.print("A");
else
  if (score >= 80.0)
    System.out.print("B");
  else
    if (score >= 70.0)
      System.out.print("C");
    else
      if (score >= 60.0)
        System.out.print("D");
      else
        System.out.print("F");
```

(a)

Equivalent

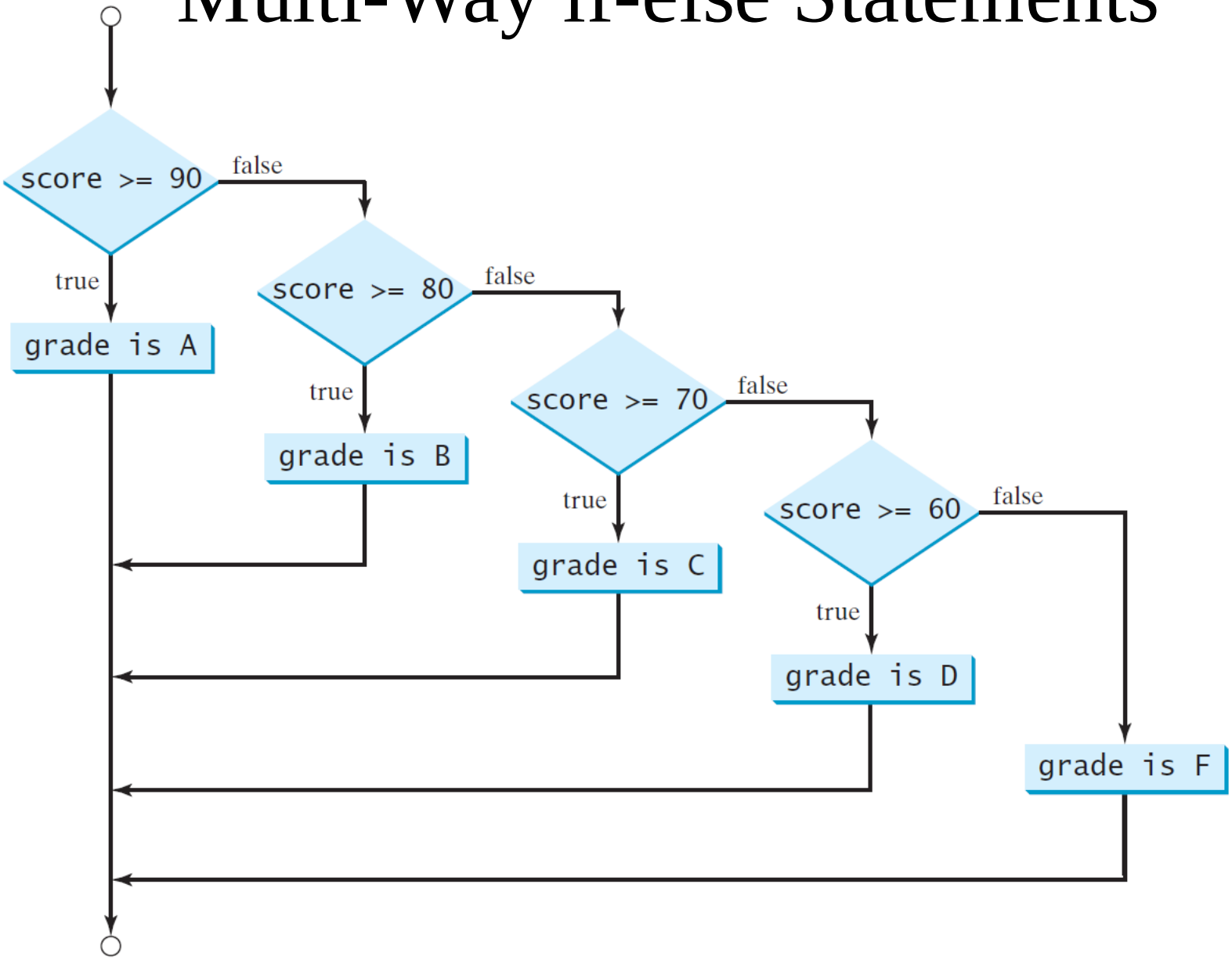
This is better

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```

(b)

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Multi-Way if-else Statements



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace if-else statement

Suppose score is 70.0

The condition is false

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace if-else statement

Suppose score is 70.0

The condition is false

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace if-else statement

Suppose score is 70.0

The condition is true

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace if-else statement

Suppose score is 70.0

grade is C

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace if-else statement

Suppose score is 70.0

Exit the if statement

```
if (score >= 90.0)
  System.out.print("A");
else if (score >= 80.0)
  System.out.print("B");
else if (score >= 70.0)
  System.out.print("C");
else if (score >= 60.0)
  System.out.print("D");
else
  System.out.print("F");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Note

The else clause matches the most recent if clause in the same block.

```
int i = 1, j = 2, k = 3;

if (i > j)
  if (i > k)
    System.out.println("A");
else
    System.out.println("B");
```

(a)

Equivalent

This is better with correct indentation

```
int i = 1, j = 2, k = 3;

if (i > j)
  if (i > k)
    System.out.println("A");
  else
    System.out.println("B");
```

(b)



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Note, cont.

Nothing is printed from the preceding statement. To force the else clause to match the first if clause, you must add a pair of braces:

```
int i = 1;
int j = 2;
int k = 3;
if (i > j) {
  if (i > k)
    System.out.println("A");
}
else
  System.out.println("B");
```

This statement prints B.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Common Errors

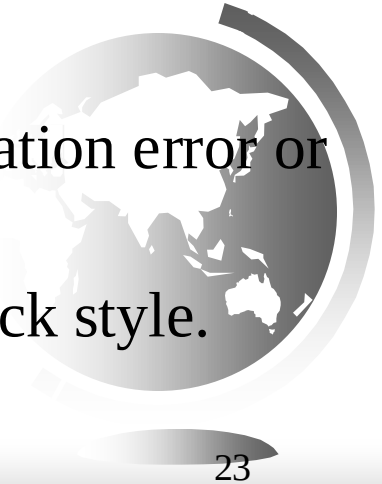
Adding a semicolon at the end of an if clause is a common mistake.

```
if (radius >= 0);
{
  area = radius*radius*PI;
  System.out.println(
    "The area for the circle of radius " +
    radius + " is " + area);
}
```

Wrong

This mistake is hard to find, because it is not a compilation error or a runtime error, it is a logic error.

This error often occurs when you use the next-line block style.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Common Errors

- Equality Test of Two Floating-Point Values

```
double x = 1.0 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1;
System.out.println(x == 0.5);
```

- it displays **false**.
- **x** is not exactly **0.5**, but is 0**.5000000000000001**.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

TIP

```
if (number % 2 == 0)
  even = true;
else
  even = false;
```

(a)

Equivalent

```
boolean even
  = number % 2 == 0;
```

(b)



Copyright © 2022 Pearson Education Ltd. All rights reserved.

CAUTION

```
if (even == true)
  System.out.println(
    "It is even.");
```

(a)

Equivalent

```
if (even)
  System.out.println(
    "It is even.");
```

(b)



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Example#1

What is the output of the following code if **number** is **14**, **15**, or **30**?

```
if (number % 2 == 0)
   System.out.println(number + " is even");
if (number % 5 == 0)
   System.out.println(number + " is multiple of 5");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Example#2

What is the output of the following code if **number** is **14**, **15**, or **30**?

```
if (number % 2 == 0)
   System.out.println(number + " is even");
else if (number % 5 == 0)
   System.out.println(number + " is multiple of 5");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: An Improved Math Learning Tool

This example creates a program to teach a first grade child how to learn subtractions. The program randomly generates two single-digit integers number1 and number2 with number1 >= number2 and displays a question such as "What is 9 – 2?" to the student. After the student types the answer, the program displays whether the answer is correct.



SubtractionQuiz

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Body Mass Index

Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing by the square of your height in meters. The interpretation of BMI for people 16 years or older is as follows:

BMI	Interpretation
BMI < 18.5	Underweight
18.5 <= BMI < 25.0	Normal
25.0 <= BMI < 30.0	Overweight
30.0 <= BMI	Obese

ComputeAndInterpretBMI



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Computing Taxes

The US federal personal income tax is calculated based on the filing status and taxable income. There are four filing statuses: single filers, married filing jointly, married filing separately, and head of household. The tax rates for 2009 are shown below.

Marginal Tax Rate	*Single*	*Married Filing Jointly or Qualifying Widow(er)*	*Married Filing Separately*	*Head of Household*
10%	$0 – $8,350	$0 – $16,700	$0 – $8,350	$0 – $11,950
15%	$8,351 – $33,950	$16,701 – $67,900	$8,351 – $33,950	$11,951 – $45,500
25%	$33,951 – $82,250	$67,901 – $137,050	$33,951 – $68,525	$45,501 – $117,450
28%	$82,251 – $171,550	$137,051 – $208,850	$68,526 – $104,425	$117,451 – $190,200
33%	$171,551 – $372,950	$208,851 – $372,950	$104,426 – $186,475	$190,201 – $372,950
35%	$372,951+	$372,951+	$186,476+	$372,951+

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Computing Taxes, cont.

```
if (status == 0) {
  // Compute tax for single filers
}
else if (status == 1) {
  // Compute tax for married file jointly
  // or qualifying widow(er)
}
else if (status == 2) {
  // Compute tax for married file separately
}
else if (status == 3) {
  // Compute tax for head of household
}
else {
  // Display wrong status
}
```

ComputeTax



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Logical Operators

Operator	Name	Description
!	not	logical negation
&&	and	logical conjunction
||	or	logical disjunction
^	exclusive or	logical exclusion

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Truth Table for Operator !

p	!p	Example (assume age = 24, weight = 140)
true	false	!(age > 18) is false, because (age > 18) is true.
false	true	!(weight == 150) is true, because (weight == 150) is false.

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Truth Table for Operator &&

p_1	p_2	p_1 && p_2	Example (assume age = 24, weight = 140)
false	false	false	(age <= 18) && (weight < 140) is false, because both conditions are both false.
false	true	false	
true	false	false	(age > 18) && (weight > 140) is false, because (weight > 140) is false.
true	true	true	(age > 18) && (weight >= 140) is true, because both (age > 18) and (weight >= 140) are true.

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Truth Table for Operator ||

p_1	p_2	p_1 || p_2	Example (assume age = 24, weihgt = 140)
false	false	false	
false	true	true	(age > 34) || (weight <= 140) is true, because (age > 34) is false, but (weight <= 140) is true.
true	false	true	(age > 14) || (weight >= 150) is false, because (age > 14) is true.
true	true	true	

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Truth Table for Operator ^

p_1	p_2	p_1 ^ p_2	Example (assume age = 24, weight = 140)
false	false	false	(age > 34) ^ (weight > 140) is true, because (age > 34) is false and (weight > 140) is false.
false	true	true	(age > 34) ^ (weight >= 140) is true, because (age > 34) is false but (weight >= 140) is true.
true	false	true	(age > 14) ^ (weight > 140) is true, because (age > 14) is true and (weight > 140) is false.
true	true	false	

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Examples

Here is a program that checks whether a number is divisible by 2 and 3, whether a number is divisible by 2 or 3, and whether a number is divisible by 2 or 3 but not both:

TestBooleanOperators



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Examples

```
System.out.println("Is " + number + " divisible by 2 and 3? " +
  ((number % 2 == 0) && (number % 3 == 0)));

System.out.println("Is " + number + " divisible by 2 or 3? " +
  ((number % 2 == 0) || (number % 3 == 0)));

 System.out.println("Is " + number +
   " divisible by 2 or 3, but not both? " +
   ((number % 2 == 0) ^ (number % 3 == 0)));
```

TestBooleanOperators



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Determining Leap Year?

This program first prompts the user to enter a year as an int value and checks if it is a leap year.

A year is a leap year if it is divisible by 4 but not by 100, or it is divisible by 400.

(year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)

LeapYear



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Lottery

Write a program that randomly generates a lottery of a two-digit number, prompts the user to enter a two-digit number, and determines whether the user wins according to the following rule:

- If the user input matches the lottery in exact order, the award is $10,000.
- If the user input matches the lottery, the award is $3,000.
- If one digit in the user input matches a digit in the lottery, the award is $1,000.

Lottery



Copyright © 2022 Pearson Education Ltd. All rights reserved.

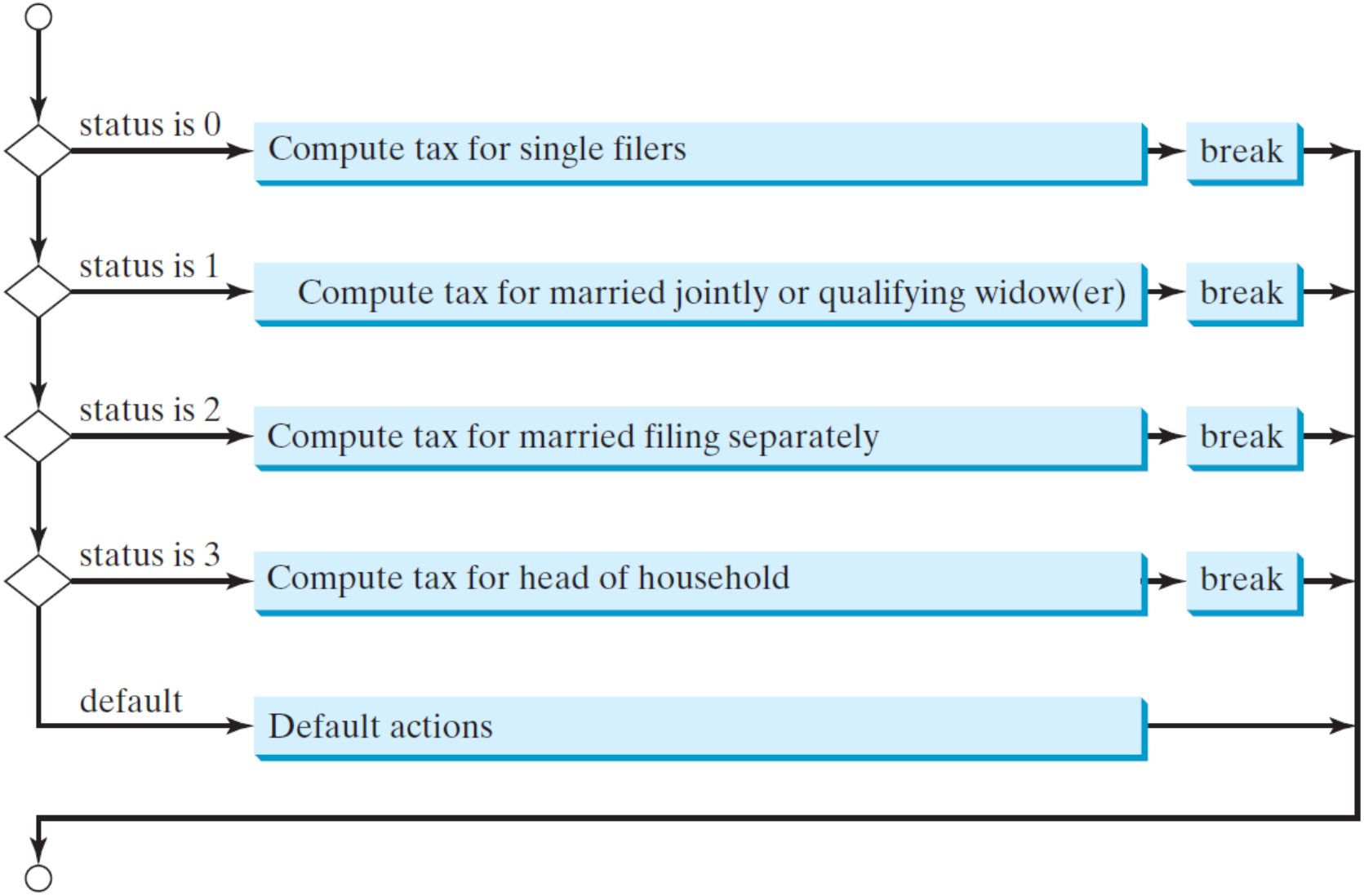
`switch` Statements

```
switch (status) {
  case 0:  compute taxes for single filers;
           break;
  case 1:  compute taxes for married file jointly;
           break;
  case 2:  compute taxes for married file separately;
           break;
  case 3:  compute taxes for head of household;
           break;
  default: System.out.println("Errors: invalid status");
           System.exit(1);
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

`switch` Statement Flow Chart



Copyright © 2022 Pearson Education Ltd. All rights reserved.

`switch` Statement Rules

The switch-expression must yield a value of char, byte, short, or int type and must always be enclosed in parentheses.

The value1, ..., and valueN must have the same data type as the value of the switch-expression. The resulting statements in the case statement are executed when the value in the case statement matches the value of the switch-expression. Note that value1, ..., and valueN are constant expressions, meaning that they cannot contain variables in the expression, such as 1 + x.

```
switch (switch-expression) {
  case value1:  statement(s)1;
                break;
  case value2: statement(s)2;
                break;
  …
  case valueN: statement(s)N;
                break;
  default: statement(s)-for-default;
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

`switch` Statement Rules

The keyword break is optional, but it should be used at the end of each case in order to terminate the remainder of the switch statement. If the break statement is not present, the next case statement will be executed.

```
switch (switch-expression) {
  case value1:  statement(s)1;
        break;
  case value2: statement(s)2;
        break;
  …
  case valueN: statement(s)N;
        break;
  default: statement(s)-for-default;
}
```

The default case, which is optional, can be used to perform actions when none of the specified cases matches the switch-expression.

When the value in a **case** statement matches the value of the **switch-expression**, the statements *starting from this case* are executed until either a **break** statement or the end of the **switch** statement is reached.

Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Suppose day is 2:

```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Match case 2

```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Fall through case 3

```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Fall through case 4

```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Fall through case 5

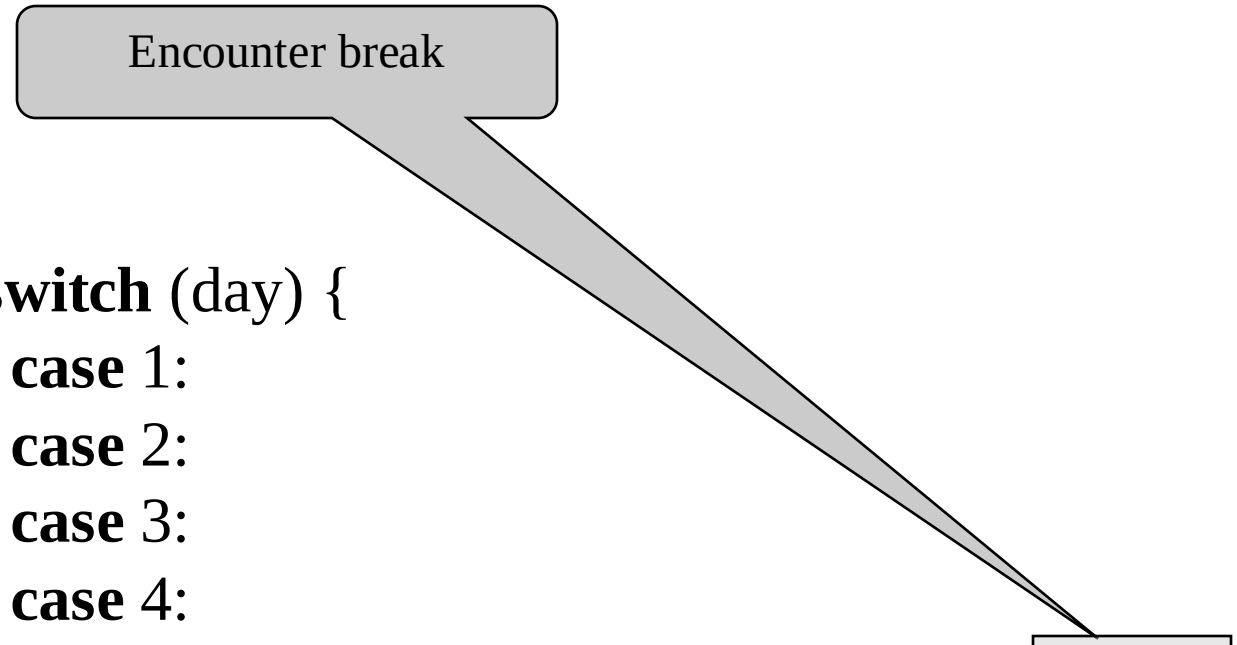
```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement




```
switch (day) {
  case 1:
  case 2:
  case 3:
  case 4:
  case 5: System.out.println("Weekday"); break;
  case 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

animation

Trace switch statement

Exit the statement

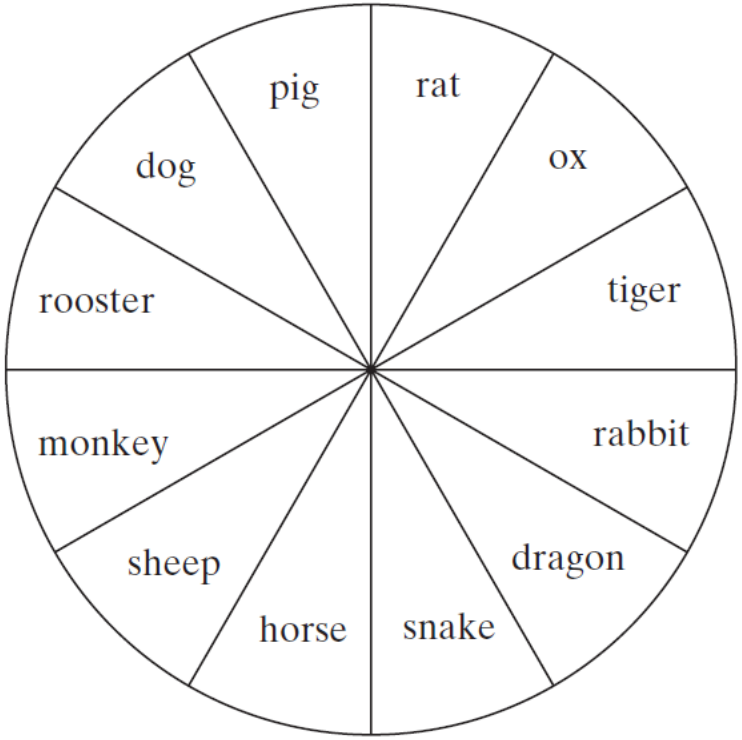
```
swit   (day) {
  ca   :
  ca  2:
  c   3:
  c  e 4:
  c  se 5: System.out.println("Weekday"); break;
   ase 0:
  case 6: System.out.println("Weekend");
}
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Problem: Chinese Zodiac

Write a program that prompts the user to enter a year and displays the animal for the year.



year % 12 =

0: monkey
1: rooster
2: dog
3: pig
4: rat
5: ox
6: tiger
7: rabbit
8: dragon
9: snake
10: horse
11: sheep

ChineseZodiac

Copyright © 2022 Pearson Education Ltd. All rights reserved.

Conditional Expressions

- *A conditional expression evaluates an expression based on a condition.*

```
if (x > 0)
  y = 1;
else
  y = -1;
```

is equivalent to

```
y = (x > 0) ? 1 : -1;
```

(boolean-expression) ? expression1 : expression2



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Conditional Operator

```
if (num % 2 == 0)
  System.out.println(num + "is even");
else
  System.out.println(num + "is odd");
```

```
System.out.println(
  (num % 2 == 0)? num + "is even" :
  num + "is odd");
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Conditional Operator, cont.

```
boolean-expression ? exp1 : exp2
```



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Operator Precedence

- **var++, var--**
- **+, - (Unary plus and minus), ++var,--var**
- **(type) Casting**
- **! (Not)**
- ***, /, % (Multiplication, division, and remainder)**
- **+, - (Binary addition and subtraction)**
- **<, <=, >, >= (Relational operators)**
- **==, !=; (Equality)**
- **^ (Exclusive OR)**
- **&& (Conditional AND) Short-circuit AND**
- **|| (Conditional OR) Short-circuit OR**
- **=, +=, -=, *=, /=, %= (Assignment operator)**



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Operator Precedence and Associativity

- The expression in the parentheses is evaluated first. (Parentheses can be nested, in which case the expression in the inner parentheses is executed first.)

- When evaluating an expression without parentheses, the operators are applied according to the precedence rule and the associativity rule.

- If operators with the same precedence are next to each other, their associativity determines the order of evaluation. All binary operators except assignment operators are left-associative.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Operator Associativity

When two operators with the same precedence are evaluated, the *associativity* of the operators determines the order of evaluation. All binary operators except assignment operators are *left-associative*.

a – b + c – d is equivalent to ((a – b) + c) – d

Assignment operators are *right-associative*. Therefore, the expression

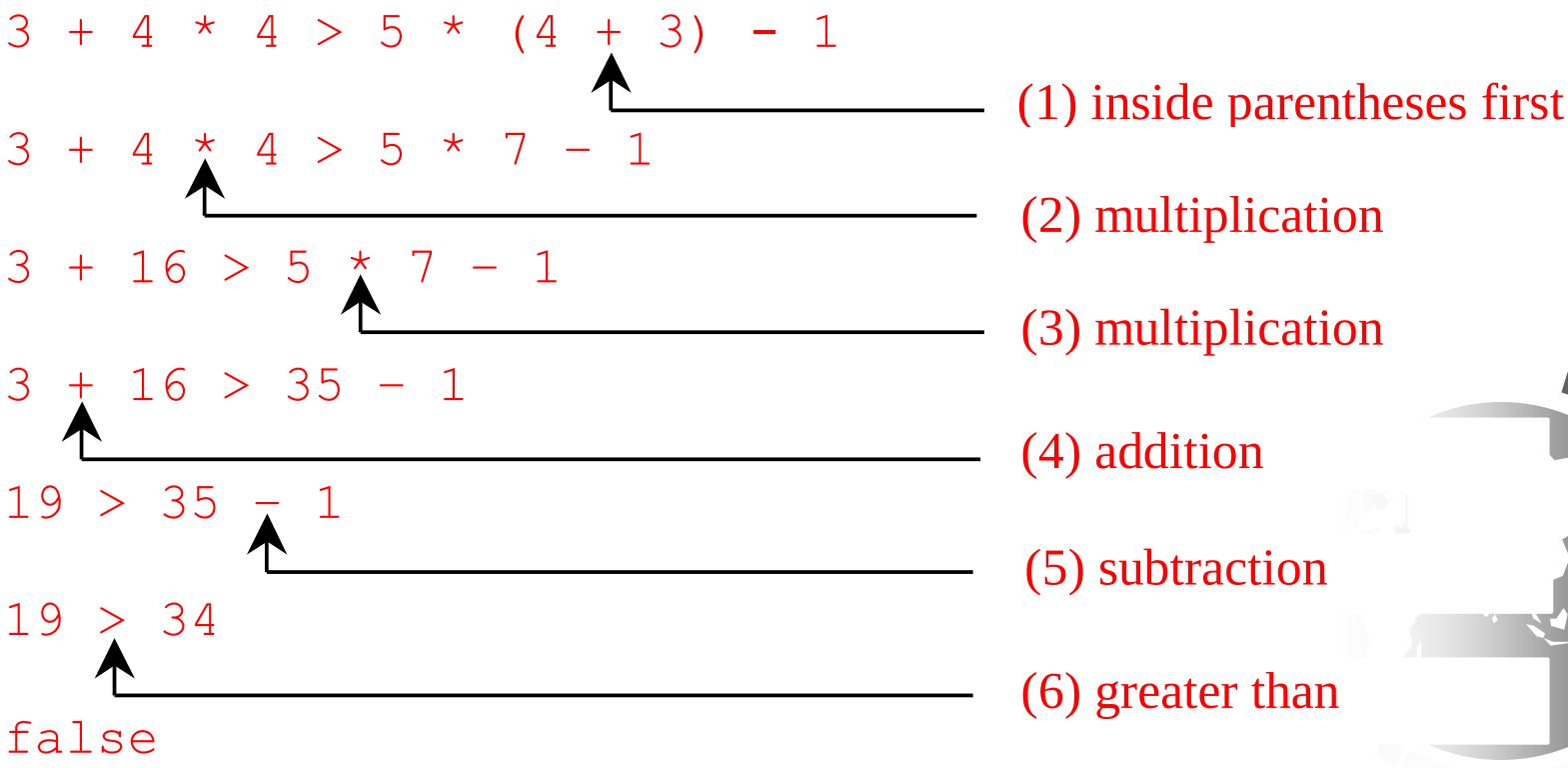
a = b += c = 5 is equivalent to a = (b += (c = 5))



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Example

Applying the operator precedence and associativity rule, the expression 3 + 4 * 4 > 5 * (4 + 3) - 1 is evaluated as follows:



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Debugging

- Logic errors are called *bugs*.
- The process of finding and correcting errors is called *debugging*.
- You can hand-trace the program (i.e., catch errors by reading the program), or you can insert print statements in order to show the values of the variables or the execution flow of the program.
- This approach might work for a short, simple program.
- But for a large, complex program, the most effective approach for debugging is to use a debugger utility.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Debugger

Debugger is a program that facilitates debugging. You can use a debugger to

- Execute a single statement at a time.
- Trace into or stepping over a method.
- Set breakpoints.
- Display variables.
- Display call stack.
- Modify variables.



Copyright © 2022 Pearson Education Ltd. All rights reserved.

Companion Website

Debugging in Eclipse

Supplement II.G, Learning Java Effectively with Eclipse



Copyright © 2022 Pearson Education Ltd. All rights reserved.